Near-Exact Filtering and Parameter Inference for Robust Location Models: Particle Filtering and Near-Exact Benchmarking for Robust State-Space Models

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1. Motivation and background

Robust filtering of a time-varying signal $\{\mu_t\}$ from heavy-tailed observations $\{y_t\}$ is central in econometrics and time-series analysis. This project focuses on reconstructing the filtering distribution $p(\mu_t \mid y_{0:t})$ of a robust state-space model using particle methods [2, 1]. Score-driven recursions [3] serve as a tractable observation-driven baseline (see references therein for broader GAS developments).

Following the notation in Density2 [5], I study a Student-t observation equation and a limit-Gaussian state innovation family. This setup is ideal to quantify: (i) how accurately particle methods reconstruct the filtering distribution and how the score-driven approximation error varies with ν, and (ii) how identification of tail-thickness weakens as $\nu \to \infty$ (near-Gaussian regime).

2. Models of interest

(A) Score-driven (observation-driven) robust location. Let

$$y_t \mid \mathcal{F}_{t-1}, \mu_t \sim t_\nu(\mu_t, \sigma),$$

with $\nu > 2$, and define the robust scaled score

$$x_t = \frac{y_t - \mu_t}{1 + (y_t - \mu_t)^2/(\nu\sigma^2)}.$$

The recursion is

$$\mu_t = \phi\mu_{t-1} + \kappa x_{t-1}, \qquad |\phi| < 1,$$

with static parameters $\theta_A = (\phi, \kappa, \sigma, \nu)$ [3].

(B) State-space (unobserved component) robust location. Consider

$$y_t = \mu_t + \varsigma_t, \qquad \varsigma_t \sim t_\nu(0, \sigma), \tag{1}$$

$$\mu_t = \phi\mu_{t-1} + \eta_t, \qquad \eta_t \sim \ell_\nu(0, \varpi), \tag{2}$$

where ℓ_ν denotes the Density2 limit-Gaussian innovation family (sub-Gaussian for finite ν, Gaussian in the limit) [5]. The filtering target is $p(\mu_t \mid y_{0:t})$.

3. Research aims and questions (core contribution)

The project has three tightly linked aims.

Aim A: Near-exact 1D benchmarking and evaluation of particle filter performance

Q(A1): In 1D model (B), how accurately do particle methods reconstruct $p(\mu_t \mid y_{0:t})$ across regimes of $(\nu, \sigma, \varpi, \phi)$ and under outliers, and what is the approximation error of the score-driven baseline?

Approach. I will use a deterministic 1D grid/quadrature filter as near-exact benchmark for $p(\mu_t \mid y_{0:t})$. Against this reference I will evaluate: (i) bootstrap PF and APF as the primary reconstruction methods, (ii) PMMH-based parameter learning, and (iii) score-driven filtering as an observation-driven baseline.

Metrics. Filtering accuracy (point-estimate error, predictive scoring rules), uncertainty quantification diagnostics, and computational efficiency.

Aim B: When does the model behave "Kalman-like" and how to operationalise the regime

Q(B1): For what ν do simpler approximations (Kalman and score-driven) become practically adequate substitutes for the exact non-Gaussian filter?

Approach. Using the 1D benchmark, I will quantify distances between the exact non-Gaussian filter and both Kalman-approximate and score-driven filtering distributions (Wasserstein and moment gaps), and define an operational threshold ν_0 above which simpler approximations are acceptable. Insights from this 1D analysis may also guide extensions to higher-dimensional settings where the grid benchmark is no longer feasible.

Aim C: Curvature/identifiability of ν and implications for inference

Q(C1): How does identifiability of ν degrade as $\nu \to \infty$, and what is the impact on MLE/Bayesian learning?

Approach. I will study: (i) profile likelihood $\ell(\nu)$, (ii) curvature diagnostics under transformed parametrisations (e.g. $\log(\nu - 2)$), (iii) Bayesian posteriors for ν under PMMH across ν_{true}. The goal is a clear statement of the "flat likelihood" phenomenon and practical remedies (reparametrisation, priors, or threshold-to-Gaussian).

Optional extension (only if core aims complete). As a secondary direction, I will assess whether model (B) admits computable finite-mixture or exact filtering approximations by investigating whether the $\ell_\nu(0, \varpi)$ state transition admits a dual process in the sense of Kon Kam King et al. [4], which would yield exact recursive filtering distributions as finite mixtures of conjugate densities. If pursued, this will be reported as an exploratory appendix, not a core deliverable.

4. Literature positioning (selected)

- **Score-driven/GAS:** robust score-driven filtering and related methodological background [3]; see references therein.
- **Non-Gaussian state-space inference:** particle filtering and particle MCMC [2, 1].
- **Density2/limit-Gaussian innovations:** sub-Gaussian innovation family, Gaussian limit, and link between score-driven and optimal filtering [5].

5. Methodology and experimental design

5.1 Implementation plan

1. **1D near-exact grid filter:** prediction/update quadrature for model (B), using the Density2 innovation law ℓ_ν.
2. **Particle methods:** bootstrap PF and auxiliary PF with diagnostics (ESS, resampling frequency, variance of $\log \widehat{p}(y \mid \theta)$).
3. **Parameter inference:**

- model (A): MLE/profile likelihood for ν;
- model (B): PMMH using unbiased PF likelihood estimates.

5.2 Evaluation

- **Parameter sweeps:** $\nu \in \{5, 10, 30, 100, 1000\}$, multiple (σ, ϖ, ϕ) regimes, with/without additive outliers.
- **Outputs:** filtering accuracy metrics, posterior summaries for ν, runtime, ESS, and variance of PF log-likelihood estimates.
- **Empirical validation:** assessment on time series consistent with the model structure, subject to data availability.

6. Data

Evaluation is conducted on time series consistent with models (A) and (B): a latent location process with robust heavy-tailed observations, where the signal-to-noise regime and tail thickness are controlled. This structure enables ground-truth assessment of μ_t and controlled evaluation of particle filter reconstruction accuracy, and is representative of a broad class of heavy-tailed time series encountered in practice.

7. Feasibility, milestones, and risks

Milestones (high level).

- **Mar–Apr:** finalise 1D grid benchmark, PF/APF diagnostics, and reproducible experiment harness.
- **May–Jun:** PMMH experiments, profile-curvature analysis for ν, and threshold-to-Gaussian diagnostics.
- **Jul–Aug:** consolidate results, empirical validation, and writing; pursue dual-process exact filter investigation only if core outputs are complete.

Key risks and mitigations.

- **Computational cost (PMMH):** mitigate via APF, variance diagnostics, adaptive particle counts, and parallel runs.
- **Weak identification for large ν:** report profile/posterior flatness explicitly and use transformed parametrisation plus informative priors.
- **Scope risk:** protect deliverables by prioritising Aims A–C; exploratory exact/computable filtering is optional.

8. Expected contribution and impact

The project will deliver: (i) a near-exact 1D benchmark for robust filtering under Density2 innovations, (ii) an evaluation of particle filter reconstruction accuracy and score-driven approximation error relative to the near-exact benchmark, and (iii) a practical characterisation of the $\nu \to \infty$ Kalman-like regime, including identifiability/curvature diagnostics and operational guidance.

References

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